

# RAHUL MITRA

## OPERATIONS RESEARCH ENGINEER PROFILE

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### PROFILE SUMMARY

Early-career operations research engineer combining NUS ISE foundations with deployable software, simulation, optimization, and analytics automation for scheduling, digital twin, process improvement, and decision-support roles.

### EDUCATION

#### National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Operations Research; Simulation; Statistics for Engineering Applications; Industrial Systems Engineering; Artificial Intelligence; Software Engineering

#### Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

### WORK AND INTERNSHIP EXPERIENCE

#### Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

#### Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Jun 2026

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.

#### Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

#### Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Built an end-to-end digital garden-plot balloting platform (admin and public-facing frontends, backend, GIS mapping, and deployment), now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and technical project management across product, engineering, and operations stakeholders; awarded the S\$20,000 Sparks Community Innovation Fund.

### SELECTED PROJECTS AND COMPETITIONS

#### Hybrid Flow Shop Digital Twin Optimizer

2026

- Modeled production scheduling as a digital twin simulation and constraint-programming optimization problem, connecting ISE methods with deployment-oriented Python tooling.

#### Volt Pulse SG - SMU Hack For Cities Finalist

Jan 2026

- Reached Top 8 of 50+ teams with an agentic AI platform for HDB energy-cost tracking using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

#### IE2110 Operations Research Graph Optimization

2024

- Applied IE2110 Operations Research I concepts from linear programming, network flow models, and nonlinear programming; implemented Jupyter SSSP/APSP graph optimization for engineering decision support.

#### NVIDIA Disaster Risk Monitoring with Satellite Imagery

2023

- Completed hands-on geospatial AI training covering satellite imagery, U-Net semantic segmentation, disaster-risk assessment, and GPU-accelerated model workflows.

### TECHNICAL SKILLS AND CERTIFICATIONS

**Skills:** Constraint Programming; Scheduling Optimization; Hybrid Flow Shop; Digital Twin Simulation; Dijkstra; SSSP/APSP; Python; Jupyter; Statistics; Simulation; Excel VBA; Decision Analytics.

- Excel 2019 Advanced - analytics automation and spreadsheet modeling.
- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

### CO-CURRICULAR AND COMMUNITY

#### Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

#### Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.